INCH-POUND
MIL-DTL-24643/71
9 October 2008

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO + 125 °C, 1000 VOLTS, TYPE LSSCF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, tin-coated. See table I for size.
Second	-	Flexible extruded silicone insulation. See table I for minimum diameter.
Third	-	Flexible cross-linked polyolefin jacket. See table I for thickness and diameter.

NOTE: Minimum bend radius of the cables are the listed mandrel diameter in table I for the cold bending test.

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TABLE I. Construction details.

Military Part No. M24643/71	Type & Size	Cold Bending Mandrel Diameter (inches)	Conductor Size AWG	Conductor Strands 30 AWG Strand Size	Conductor Resistance	Insulation Diameter (inches)	Insulation Resistance (megohms)	Jacket Thickness (inches)	Jacket Diameter (inches)	
						(min.)		(min. avg.)	(min.)	(max.)
-01UN	LSSCF-3	0.75	16	26	4.624	0.110	500	0.012	0.135	0.150
-02UN	LSSCF-4	1.00	14	41	2.88	0.155	500	0.012	0.180	0.195
-03UN	LSSCF-10	1.25	10	105	1.14	0.195	500	0.017	0.230	0.245
-04UN	LSSCF-16	1.50	8	168	0.729	0.270	500	0.022	0.320	0.340
-05UN	LSSCF-26	1.75	6	259	0.459	0.315	250	0.022	0.365	0.385
-06UN	LSSCF-42	2.25	4	420	0.289	0.370	200	0.032	0.440	0.460
-07UN	LSSCF-66	2.50	2	665	0.184	0.435	150	0.042	0.525	0.555
-08UN	LSSCF-105	3.25	1/0	1045	0.115	0.555	125	0.047	0.655	0.685
-09UN	LSSCF-133	3.50	2/0	1330	0.0922	0.605	100	0.047	0.705	0.735
-10UN	LSSCF-165	4.00	3/0	1672	0.0731	0.680	100	0.047	0.780	0.820
-11UN	LSSCF-210	4.25	4/0	2109	0.0580	0.730	90	0.047	0.830	0.870
-12UN	LSSCF-250	4.50	250 MCM	2449	0.0491	0.830	80	0.047	0.930	0.970
-13UN	LSSCF-300	4.75	300 MCM	2989	0.0409	0.860	75	0.047	0.960	1.010
-14UN	LSSCF-350	5.00	350 MCM	3458	0.0354	0.890	75	0.047	0.990	1.040
-15UN	LSSCF-400	5.25	400 MCM	3990	0.0310	0.960	70	0.047	1.065	1.115

REQUIREMENTS:

Qualification required.

EXAMINATION AND TESTS:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table I
Voltage withstand, conductor to ground (volts, root mean square, min.)	5000
Insulation resistance, conductor to ground (megohms/1000 feet, min.)	See table I
Conductor continuity	No Failure
Group A:	
Visual and dimensional	No Failure
Group B:	
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Cold bending, cable (tested at -20 °C, mandrel diameter iaw table I)	No Damage
Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	700

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Elongation (percent, min.)	250
Jacket (cable)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	200
Tear (pounds per inch thickness, min.)	35
Jacket (durometer – Type A hardness, min./max.)	60/75
Group C:	
Physicals (aged)	
Insulation	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Jacket (cable)	
Air oven	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No Failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (jacket) (cycles, min.)	75
Group D:	
Flame propagation (cable)	No Failure
Qualification Inspection:	
Qualification inspection shall include basic electricals, groups A, B, C, and D, plus the following:	
Aging and compatibility (cable) (125±5 °C)	No Failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Insulation	0.2
Smoke index, max.	
Jacket	25
Insulation	35

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Toxicity index, max.	
Jacket	5
Insulation	1.5
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Electrical moisture absorption	No Failure
Weathering (jacket)	No Failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSSCF - 3 through 66	1000
LSSCF - 105 through 400	500

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-118)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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